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FACULTY OF INFORMATION TECHNOLOGY

FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

DEPARTMENT OF INTELLIGENT SYSTEMS

ÚSTAV INTELIGENTNÍCH SYSTÉMŮ

DETECTING AI-GENERATED FAKE REVIEWS IN ON-LINE PLATFORMS

DETEKCE FALEŠNÝCH RECENZÍ GENEROVANÝCH AI NA ONLINE PLATFORMÁCH

BACHELOR'S THESIS

BAKALÁŘSKÁ PRÁCE

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BRNO 2026

Abstract

This thesis aims to implement a classifier for detecting fake reviews generated by artificial intelligence on shopping platforms, focusing on Slovak reviews. The thesis analyzes the impact of fake reviews on the reputation and sales of products and services. A synthetic dataset is created by scraping Slovak reviews from Booking.com, generating another set of reviews via a language model, and combining them. Then, the thesis presents a method for classifying these reviews. **[[It does this by this and that]]. [[The results are this and that]].** In addition to this, the thesis also compares and discusses the results of simpler methods, such as some traditional machine learning methods.

Abstrakt

Táto práca sa zameriava na implementáciu klasifikátora na detekciu falošných recenzií na shopping platformách, so zameraním na slovenské recenzie. Práca analyzuje reputačný a finančný dopad falošných recenzií na produkty a služby. Práca taktiež vytvára syntetický dataset zbieraním slovenských recenzií z Booking.com a ich kombináciou s recenziami vygenerovanými jazykovými modelmi. **[[Robí to tak a tak.]] [[Výsledky sú to a to]].** Okrem toho, práca poskytuje porovnanie s výberom jednodušších metód, ako napríklad zopár klasických techník strojového učenia.

Keywords

Fake reviews, Machine learning, Artificial Intelligence, Text classification, Natural language processing, Python, Large language models

Klíčová slova

Falošné recenzie, Strojové učenie, Umelá inteligencia, Klasifikácia textu, Spracovanie prirodzeného jazyka, Python, Veľké jazykové modely

Reference

LACKO, Igor. *Detecting AI-Generated Fake Reviews in Online Platforms*. Brno, 2026. Bachelor's thesis. Brno University of Technology, Faculty of Information Technology. Supervisor Ing. Anton Firc

Detecting AI-Generated Fake Reviews in Online Platforms

Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Ing. Anton Firc. I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis. I have used large language models (ChatGPT, Gemini) for the language revision of the thesis. I have used Github Copilot to assist with the implementation part of the thesis.

.....

Igor Lacko
January 17, 2026

Acknowledgements

[[meow]]

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Chapter 1

Introduction

Fake reviews have been a problem for online platforms since their inception. Reviews are often the first thing someone reads when considering buying a product or using a service. As such, they are one of the biggest influences in it's reputation, which in turn influences it's sales. Moreover, people may not even find out about a product if it's reputation is damaged by fake reviews, or in a opposite example, if a product is not artificially propped up. It's no wonder that people or companies often try to take advantage of this, whether it's for their advantage via faking positive reviews on their products or to the disadvantage of potential competitors via faking negative reviews on their products. The „traditional“ way of faking reviews in the past has been to manually write them, or pay for someone else to do it, which is time consuming and costly due to requiring manual human labor. However, with the rise of generative transformer language models, such as *GPT*, and the ease of access to them, fake reviews can now be mass produced for cheap. The detection of fake reviews, explored in detail in is a continuously developing field. This is due to multiple factors: human written fake reviews differ from AI generated ones, and even among the latter, as language models keep evolving, the methods for detecting their output have to as well. As humans have been shown not to be consistently able to correctly classify reviews as real or fake, as well as manual classification not being feasible from a time perspective, there are multiple benefits to automating this process. A common way to do this is by using machine learning (ML) methods, whether via more „traditional“ ML models such as *Naive Bayes*, *Logistic regression* or *Support vector machines (SVM)*, or more recently via utilizing language models. Language models can be utilized for the classification task itself, or be used to extract features such as word embeddings or sentiment. In addition to ML techniques, another common method is using stylometric features such as text length or writing style. However, these approaches, mainly the machine learning approach, introduce another challenge, which is collecting relevant testing and training data. Existing datasets are few and far between, overly represent the hospitality domain and are mostly in English. To my best knowledge, no annotated dataset of Slovak reviews exists as of now. This work aims to create a synthetic dataset in the Slovak language by combining existing reviews written in Slovak, using a generative model to generate the „fake half“ of the dataset, and train a classifier on the created dataset for the purpose of detecting reviews generated by language models. The classifier takes word embeddings extracted using a pre-trained BERT-like model for the Slovak language in addition to various stylometric features and other textual features, such as TF-IDF. The text of this thesis is structured as follows: **Chapter 2** describes the necessary AI theory needed as the base for the other chapters. Firstly, **section 2.1** describes the general architecture of transformer based language models

and how they work. Secondly, [section 2.2](#) describes encoder models and why they are commonly used for text classification. Lastly, [section 3.3](#) describes generative decoder models. Chapter 3 explores the problematic of fake reviews and the impact they have. [Chapter 4](#) explores common methods and approaches for detecting fake reviews. Firstly, [section 4.1](#) summarizes various features typically used in fake review detection, such as sentiment or text simplicity. Secondly, [section 4.2](#) summarizes traditional ML methods that can be used. Lastly, [section 4.3](#) explores how transformer-based language models can be used, for classification itself and extracting features. Starting off the practical part of the thesis, [Chapter 5](#) describes the dataset used, while [section 5.1](#) describes existing datasets as inspiration, [section 5.2](#) describes where the used data comes from, how it was collected, it's legal aspects and characteristics. Then [section 5.3](#) describes how the data was split into training, validation, and test sets. [Chapter 6](#) describes the architecture and design of the model implementation itself. [Chapter 7](#) then describes how the model was evaluated. Concretely, [section 7.1](#) describes the evaluation method in detail, then [section 7.2](#) summarizes the results, and the chapter is then concluded by discussing the results, possible improvements and practical applications in [section 7.3](#). Finally, [Chapter 8](#) then concludes the thesis.

Chapter 2

Transformer based language models

Most modern language models are based on the *Transformers* architecture, which was invented by the 2017 paper *Attention is all you need* [21].

2.1 The transformers architecture in general

Language models split the input text into **tokens**, which are one part of a sentence processed by the language model. They are usually words, but they can be parts of words, or even characters. Language models have a vocabulary of tokens they „understand“, where each token has a unique numeric identifier. These identifiers are what the models actually work with. Modern language models can have very large numbers of tokens in their vocabulary, for example OpenAI’s *GPT-4* (2023) model had around 100 000 tokens in it’s vocabulary, while *GPT-4o* (2024) almost doubled that number[24]. Another important term when it comes to LLMs is the **context window**. This means the maximum amount of tokens that the model can process at the same time. Figure 1, taken from the aforementioned *Attention is all you need* shows the architecture of a transformer model. In it, we can see that the model is split into 2 main parts. These are called the *Encoder* and the *Decoder*, respectively. In essence, the encoder block captures the meaning of the input through embeddings and attention, while the decoder processes them and later converts them into probabilities for each token. Both the encoder and decoder block are composed of a stack of encoders/decoders respectively. Despite the original transformers architecture having both these parts, most models usually only have one. *Encoder-only models* are used for a range of NLP (Natural Language Processing) tasks such as text classification, sentiment analysis, or masked language modeling¹. Perhaps the most known encoder only model is *BERT* (Bidirectional Representation From Transformers), introduced in [4] and used as a foundation for further encoder only models. *Decoder-only models* are used mainly for text generation and are the underlying engine under popular chat bots such as Chat-GPT, which uses the *GPT* model (Generative Pre-Trained Transformer), introduced in [19]. Models that have both parts are called *Encoder-Decoder models* and are used for text-to-text tasks such as translation or text summarization.

¹masked language modeling – process where the model is given a input sentence with missing („masked“) parts and tries to fill in those parts

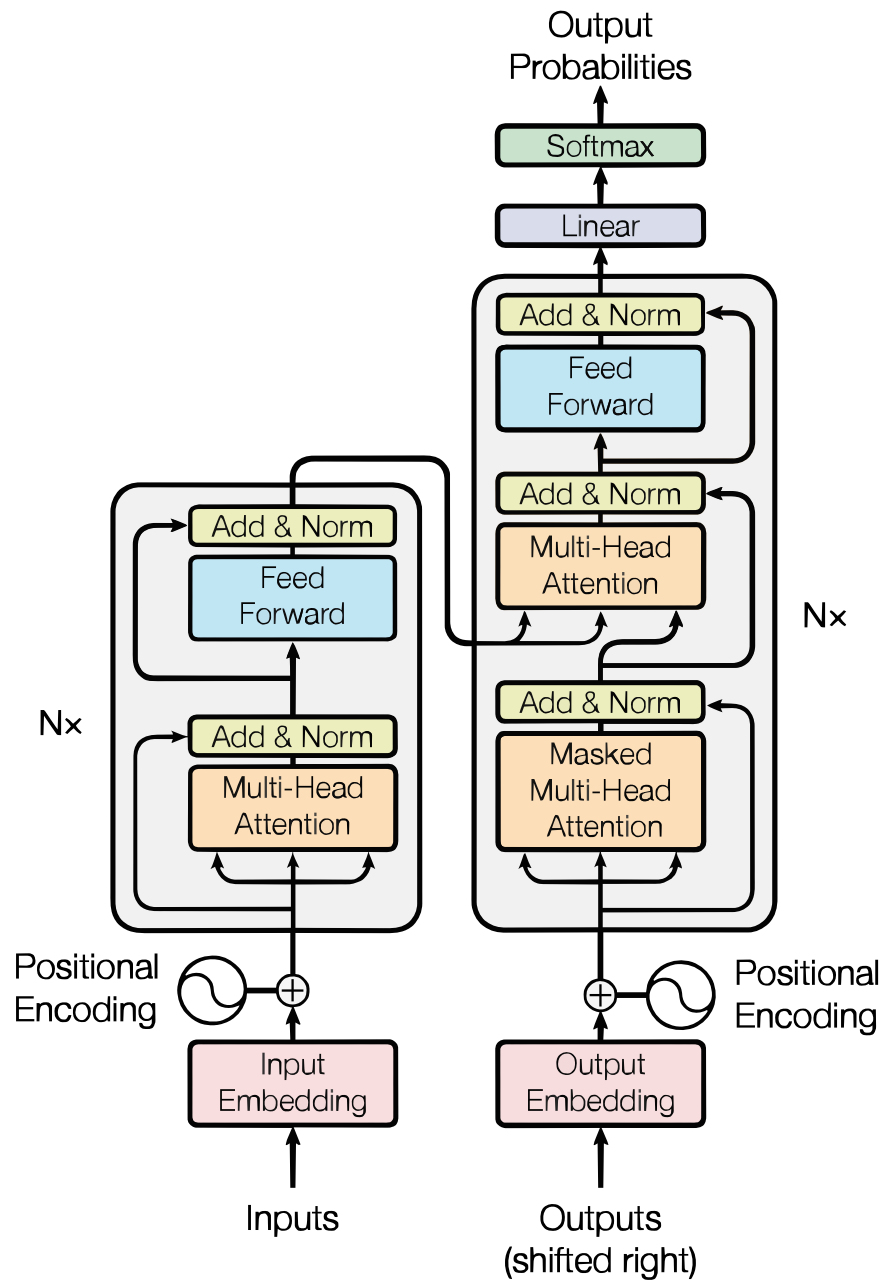


Figure 2.1: The transformers architecture.

2.2 Word embeddings and attention

What makes transformers-based models so effective is the attention mechanism. Like language models before them, transformers use *word embeddings*, which in essence are high dimensional vectors which numerically represent each input token (1 token = 1 embedding), where their values represent the semantic meaning of the text in the high-dimensional space. A well-known example are the words „king“ and „queen“. If you subtract the embedding for „man“ from „king“ and add „woman“, you get a vector that is very close to „queen“ in the high-dimensional space. One of the best known approaches used to create word embeddings was *word2vec* [13], introduced at Google in 2013, which uses a neural network. These embeddings are learned, and therefore, fixed representations of the input. Another classic example which represents what this means is the word „bank“. Since the embeddings are static, the embedding for this word will be the same for the sentences „They robbed a bank“ and „They were sitting on the river bank“. What attention does is modify the embeddings as they pass through the transformer, based on the words around them and giving more weight to tokens that may be more relevant to the currently processed one, allowing the model to *attend* to them with more importance. In other words, it gives *context* to the embeddings, so a transformers model will use different embeddings for the word „bank“ in the aforementioned examples.

2.3 Decoder models

Decoder-only models are the standard for generative NLP tasks. They work in a auto-regressive manner, by taking in the input text (the prompt) and predicting the next most likely token to appear after it. The given token is then generated, and appended to the prompt. The prompt with the generated token is then used as the input for the next cycle. The model does this until it generates the entire output.

Predicting the next token

In each cycle, as a prompt passes through a decoder, the model outputs a long list of scores (the size of it's vocabulary, which is the list of tokens „it knows“) called *logits*. These scores are normalized so that they are all positive and add up to 1 using a function called *softmax*[1]. The model then chooses the token with the highest probability as it's output. The softmax equation, taken from [23] is defined as:

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}$$

where $\mathbf{z}$ is the vector of logits and z_i is the i th logit.

System prompts

Of course, only this approach is not enough for example to get chat bots to generate concise answers which look like answering a question like most modern models do. This is why language models have additional *system prompts*, which serve as a static set of instructions or general guidelines to the model and is taken as an input alongside the user prompt. For example, a chat bot like Chat-GPT may have a system prompt which looks like: „You are a helpful chat bot which talks to people and answers their questions. The following text is

the user’s input.“. Then, the user prompt would be concatenated to this to produce the whole prompt.

Temperature

Since decoder models predict the next most likely token, they should theoretically, for the same input always give the same output (this isn’t practically always true since other hyper parameters like *top_p* or *top_k* which influence how the top results are sampled can also affect randomness [3]). However, sometimes we want the model to show non determinism, whether it is to appear „more human“, or to give a wide range of various results when generating synthetic data. *Temperature* is a parameter mostly between 0 and 2 (although some platforms/models have a different range similar to this) which controls how random the model’s output is. This works by dividing each logit by the temperature when computing the softmax equation[23], which results in higher differences between the probabilities with lower temperatures.

2.4 Encoder models

Due to the encoder component being designed to capture the understanding of the input text, encoder models are used for tasks which require this. The input of the encoder are vectors of embeddings created from the input sentence (for each token). The output is then another vector with context of the input sentence „baked“ into it using the attention mechanism. In the standard transformers architecture, this output is then passed into the decoder block. However, there is no decoder in modern BERT like models, so the usage of the output can vary. For example, the BERT paper [4] describes several ways to use the output vectors. The model was originally trained on masked language modeling. 15% of tokens were randomly chosen for masking. The tokens were replaced with a special *[MASK]* token with 80% probability, a random token with 10% probability and stayed unchanged with 10% probability. The output of the final encoder which corresponds to the position of the masked token is ran through a softmax, and the word with the highest probability is chosen as the „original“ word which was masked before. For classification, the model always has another special *[CLS]* token as the first token of the sentence. This token represents the „aggregate“ representation of the input, and is used as the input for the final layer of the model. This final layer can be task-specific, and is often called the *classification head*, which can be task-specific (as the base model does not have this), but is usually another neural network [14].

Chapter 3

Fake reviews

Fake reviews pose a huge problem to both customers and sellers. They reduce the trust of customers in online platforms and thus have an inherently negative commercial impact on online available products. **The following section** discusses the importance of reviews and the possible impacts of fake reviews.

3.1 The influence of fake reviews

Fake reviews can have various negative effects. For example, a customer may not buy a product because of negative reviews associated with it. This is obviously unjust if there is a disinformation campaign against the product. On the other side, a product may get higher sales due to being artificially propped up by positive reviews, as a tactic of the seller to gain a market advantage over the competition.

Influence on customer purchasing behavior

Perhaps the biggest factor in deciding whether to buy something is its reputation. That is why reviews exist, to present that information to people to better decide. The value of this is rooted in the presumed truthfulness of the reviews [5], which spammers¹ take advantage of. This doesn't only impact products ordered from the internet. [22] shows that around 93% of consumers say that online reviews influence their choice. More recently, [18] found that more than 70% of consumers read business reviews regularly when searching for local businesses and another 26% do so occasionally. Fake reviews can impact customer purchasing behavior in several ways. Firstly, a product with positive fake reviews can lead customers to buy it more, which can be harmful as fake reviews don't provide anything truthful regarding the value of the product. On the other side are negative reviews, which can cause consumers to distrust the product and also the seller [9]. Lastly, the knowledge of fake reviews existing itself can lead customers to distrust the review system itself and doubt all reviews [25], which can lower their purchasing intentions, indirectly harming the sales of online products or services.

Commercial impact of reviews in general and fake reviews

As the prior section implies, reviews can influence consumers purchasing behavior. This, in turn, can have real financial impacts on the reviewed products. Several studies show

¹spammer — a person who posts fake reviews [15]

correlation between review ratings and sales. For example, [11] shows that a 1 star increase in Yelp rating leads to around a 9% increase in revenue, as an example of what may lead spammers to generate untruthful reviews. Considering only fake reviews, a 2021 study by the World Economic Forum [12] found that the impact of fake reviews on online spending is around 152 billion. They based this on the total influence of reviews, which, according to a study they cite is was 3.8 trillion in 2021 and their finding on average percentage of fake reviews on e-commerce sites, which they found to be 4%. This study also says that the eventual culling of fake reviews happens in general around after 100 days. This shows that fake reviews may even be a low-risk/high-reward situation for businesses who try to promote themselves throughout them, since the considerable gains outweigh the comparatively low risk. More recently, [20] gathered statistics concerning fake reviews for 2025. They estimate that fake reviews cost customers more than 700 billion in 2025, that around 30% of online reviews are considered fake, and that more than 80% of customers encounter them. Interestingly, they found that the number of fake reviews grows about 12% faster than the number of other reviews, which may indicate increased AI usage for fabricating fake reviews. Another interesting thing they found is that 46% of the reviews considered fake have been positive in sentiment (5 stars). This correlates with other studies, such as [5] which also attribute overly positive sentiment as a possible indication of fake reviews.

3.2 Generating fake reviews using AI

In the past, fabricating fake reviews was usually done by manually writing them or buying them. Several earlier studies, such as [7] show that there exists a market for fake reviews. As the cited study says, a popular communication channel for procuring fake reviews were private Facebook groups. Nowadays, malicious actors have another method of obtaining fake reviews – using generative language models. There are many options here, such as using available APIs (for example OpenAI’s) for text generation, or just using a pre-trained generative model from the Hugging Face hub for review generation, or even just opening a language model web interface and asking to generate a review. [8] analyzed the frequency of AI-generated google reviews. They found that the rate nearly doubled from 2023 to 2024, and by 2024 almost one in five reviews on google is AI-generated. In addition to that, the quality of AI-generated reviews may be often higher than manually fabricated ones. Several studies have shown that humans have trouble distinguishing AI-generated reviews from real ones, often being no better than a random guesser (around 50%). Furthermore, studies discussing manually fabricated fake reviews talk about the inability of spammers to express true sentiment, which distinguishes their reviews from true ones. Large generative models whose training data includes reviews may not have this limitation. Another thing to consider is the ease of generating fake reviews. Manually writing fake reviews is time consuming, and the result is small in quantity. Just an online LLM can easily output 10 reviews in a single prompt. A python script controlling a loaded model for inference can output thousands in a tiny fraction of the time it would take to write them manually depending on the spammer’s hardware.

Chapter 4

Detecting AI generated reviews

4.1 Detection using non-ML methods

4.2 Detection using traditional machine learning methods

4.3 Detection using language models

Chapter 5

My method of detecting AI generated reviews

Chapter 6

Dataset

As mentioned in the introduction, to my best knowledge there is no annotated Slovak dataset of reviews. For this reason, a new dataset was collected using reviews of Slovak hotels on Booking.com.

6.1 Choice of the data source

Part of the reason why Booking.com was chosen is due to inspiration by previous datasets used in fake review detection. The hospitality sector dominates the percentage of the used datasets, per [15] and [6]. Figure 2, taken from [6] and recreated into a svg shows the distribution of the datasets used from 2021 to 2023:

Domains in fake review detection datasets from 2021 to 2023

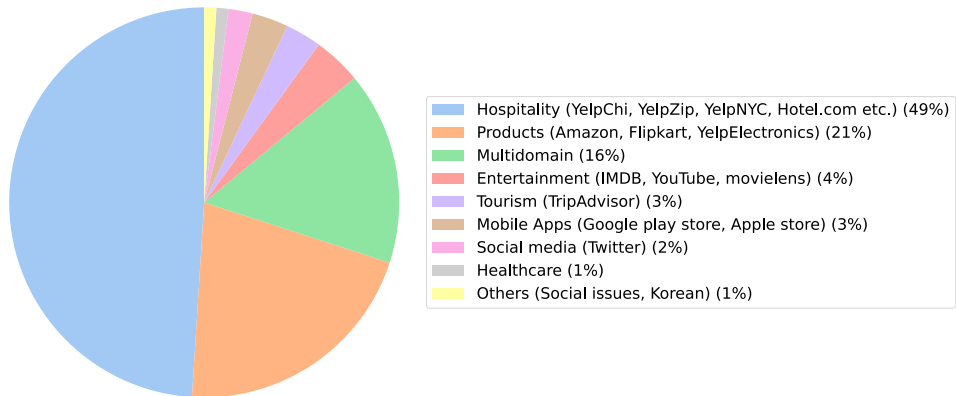


Figure 6.1: Distribution of dataset domains used in fake review detection from 2021 to 2023

Booking.com represents the largest source of reviews of hotels all over Slovakia, with options to filter by language to use only reviews in Slovak. The website can be easily parsed, and the reviews are structured into 2 parts (the „good part“, or „what did you like?“, and the „bad part“, or „what did you dislike?“) which is more complicated than 1 review = 1 text, but allows for multiple variants (use a review as one sample or as 2 samples).

6.2 Collecting data

The process of scraping

A total of 10064 reviews were scraped from the Booking.com website. The scraper always navigated to the website and searched for hotels in Slovakia. Then it would go hotel by hotel, filter the reviews by language to Slovak, and scrape each review from a hotel which it didn't scrape already (known by URL). The scraper would extract the hotel name (the individual CSV files were saved by their hotel name for filtering already scraped hotels), the positive content, negative content and rating and save them into a CSV.

Scraping permission and legal aspects of the used data

Since web scraping is a legally gray area, and Booking.com's terms and conditions [2] disallow any sort of monitoring, scraping and crawling, permission was asked from their data privacy office. They granted it under the conditions that the collected data is anonymized, the review content is not referred to, and the data is not shared. Per their request, the dataset does not contain any personal data, and it's content is not published other than aggregate statistics.

6.3 Dataset characteristics

This section describes the non synthetic part of the dataset, it's characteristics and statistics. The dataset contains 10064 rows, each of which is can be defined as a quadruple of $(rating, content_{good}, content_{bad}, content_{unified})$ where the last is just the two content parts concatenated with a whitespace between them if both are present (else it's identical to the present part). [Table 1](#) describes how many of the scraped reviews contain only positive and negative content, and how many contain both.

Content type	Count
Total reviews	10064
Only positive content	2587
Only negative content	370
Both positive and negative content	7107

Table 6.1: Statistic of content type presence in the reviews.

From the table we can see that about three quarters of the dataset have both negative and positive content. Out of those that don't, there are almost 7 times as many reviews with only positive content than reviews with only negative content. From this, it comes as no surprise that the dataset heavily skews towards positive ratings. Booking.com uses ratings in the interval $< 1, 10 >$ and [figure 3](#) shows that the mean rating is 8.24, and if we go by discrete bins, reviews with a rating of 10 are by far the most common. This might be due to Booking.com pushing highly rated hotels up as default results in a location.

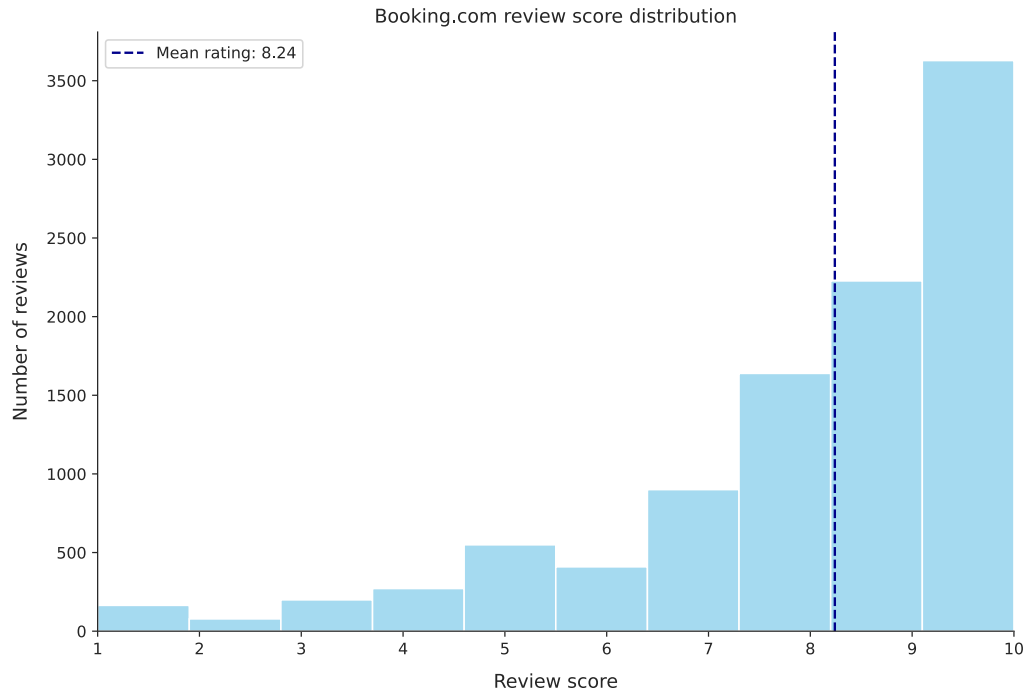


Figure 6.2: Distribution of ratings in the dataset.

When it comes to the text length, [figure 4](#) and [table 2](#) show that both the positive and negative parts of the reviews are mostly around 110 characters (in median), with a few outliers skewing the mean to the right, but both parts don't significantly differ from each other.

Content type	Average length	Median length	Max length	Min length
Positive	148.65	117.00	1988.0	1.0
Negative	159.18	106.00	1998.0	1.0
Total	262.15	195.00	2810	96

Table 6.2: Statistics of review lengths for different content types.

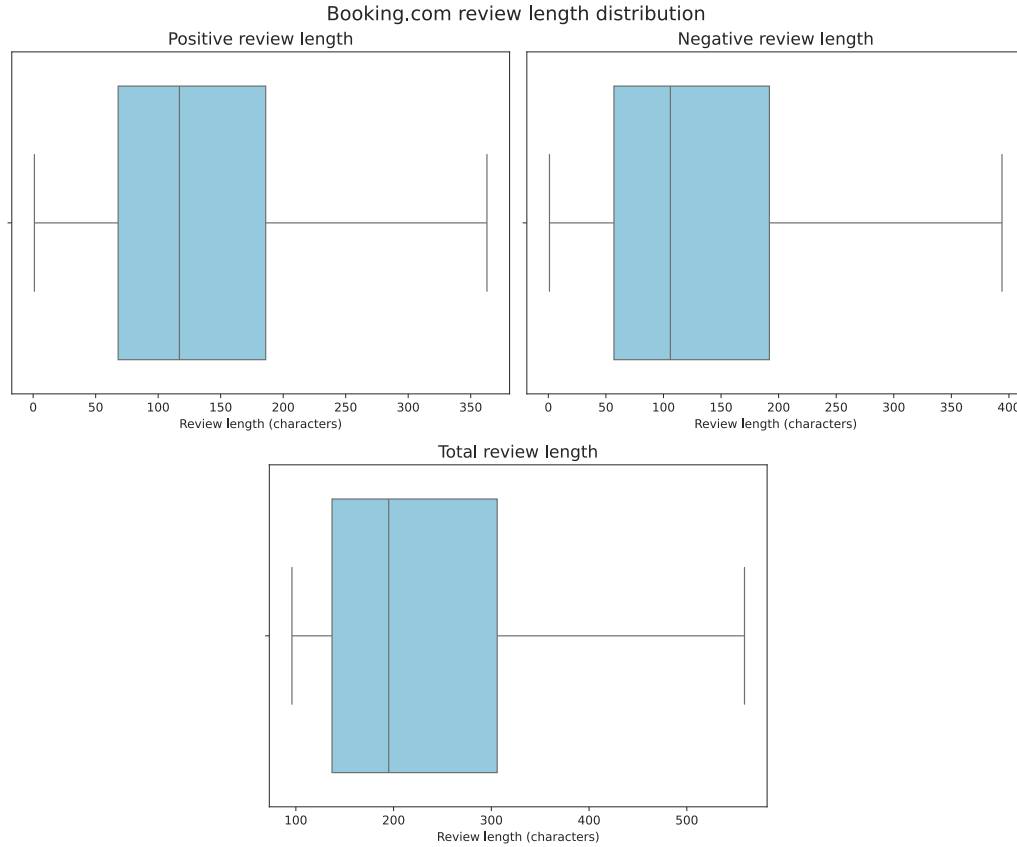


Figure 6.3: Distribution of review lengths in the dataset.

6.4 Generating synthetic data

Datasets used in the past have been constructed by methods such as crowdsourcing [15] or some other form of asking people to manually fabricate fake reviews. Another commonly used method was by using a filtering algorithm. One of the oldest datasets for fake review detection [16] collected hotel reviews from Yelp, which has its own filtering algorithm but doesn't delete reviews labeled as spam, but puts them in a separate list. However, the first method makes it difficult to gather large amounts of data, while the second labels reviews based on a non transparent algorithm. Another option is to create a new, synthetic dataset based on the „real“ one which contains scraped reviews. This involves using a generative language model to generate reviews and annotating them as fake. [5] which used GPT-3 and [10] which used various models have found success with this method. Its benefit is that it allows the generation of high quality annotated data very quickly, however we still need the „real“ part of the dataset.

Model choice

mistral-sk-7b, a version of the Mistral-7B-v0.1 language model fine-tuned for the Slovak language [17] was chosen to generate synthetic reviews. **[[This is due to...]]**

Prompts

Since mistral-sk-7b is a base model and not an instruct model, which means it's not tuned to follow instructions but rather predict the next token to appear in the text, we can't use a prompt like „*Generate a review of a hotel in Slovak...*“. Instead, the model was always given 5 examples of reviews from the dataset and was left to fill in the last one. So an example prompt looked like this:

```
Hotelová recenzia 1: <Example 1>
Hotelová recenzia 2: <Example 2>
...
Hotelová recenzia 6:
```

so the model filled in the sixth review.

Model parameters and generation

The model used variable *temperature* and *repetition_penalty* parameters, where the first was a random number from a uniform distribution on $< 0.3, 0.7 >$ (inspired by [5]), while the latter was also from a uniform distribution, but on $< 1.0, 1.2 >$. The max token size for the output was set to 512 so that it could produce outliers. From the model output, only reviews that could be truncated at the next indicator of a new review („*Hotelová recenzia*“) were taken as valid samples to filter out samples which were cut off and unfinished.

6.5 Training, validation, and test splits

Chapter 7

Implementation

Chapter 8

Evaluation

8.1 Evaluation process

8.2 Results

8.3 Discussion and practical applications

Chapter 9

Conclusion

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